

Ad Targeting Systems: Dataset Overview

The dataset tracks user impressions and their binary reaction (click/no-click) to an advertisement. The primary objective is a Supervised Classification task: predicting the likelihood of a user clicking on an ad based on their demographic and contextual data.

Column Name	Data Type	Context / Description
id	Identifier	Unique user ID. (Should be dropped or treated as a unique categorical feature).
full_name	Identifier	User placeholder name (Can be dropped).
age	Numerical	User's age. (Contains missing values).
gender	Categorical	User's reported gender (Male, Female, Non-Binary). (Contains missing values).
device_type	Categorical	Device used by the user (Desktop, Mobile, Tablet). (Contains missing values).
ad_position	Categorical	Placement of the ad on the page (Top, Bottom, Side). (Contains missing values).
browsing_history	Categorical	Category of the website content being viewed (Shopping, News, Education, Entertainment, Social Media). (Contains missing values).
time_of_day	Categorical	Time of day the ad was shown (Morning, Afternoon, Evening, Night). (Contains missing values).
click	Target (Binary)	The outcome: 1 for Click, 0 for No Click (The primary target variable for prediction).